

Title : System Design

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Submitted to-

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Chapter 7: System Design

7.1 System Architecture Design

The system architecture of the **IoT-Based Smart Patient Turning and Bed Monitoring System** is designed using a layered approach to ensure modularity, scalability, and real-time responsiveness.

The architecture consists of the following layers:

1. Sensing Layer

This layer is responsible for collecting real-time physical data from the patient and environment.

- **Pressure Sensors (FSR / Load Cell):** Placed under different sections of the mattress to detect body pressure distribution.
- **DHT22 Sensor:** Measures temperature and humidity around the patient.

These sensors continuously capture raw analog/digital signals.

2. Processing Layer

- The **ESP32 microcontroller** acts as the core processing unit.
 - It performs:
 - Sensor data acquisition
 - Data filtering and preprocessing
 - Time-based immobility tracking
 - Decision-making for alert triggering
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3. Communication Layer

- ESP32 uses **Wi-Fi module** to transmit data.
- Data is sent to cloud platforms such as:
 - Firebase
 - Blynk IoT

This layer ensures real-time data availability for remote monitoring.

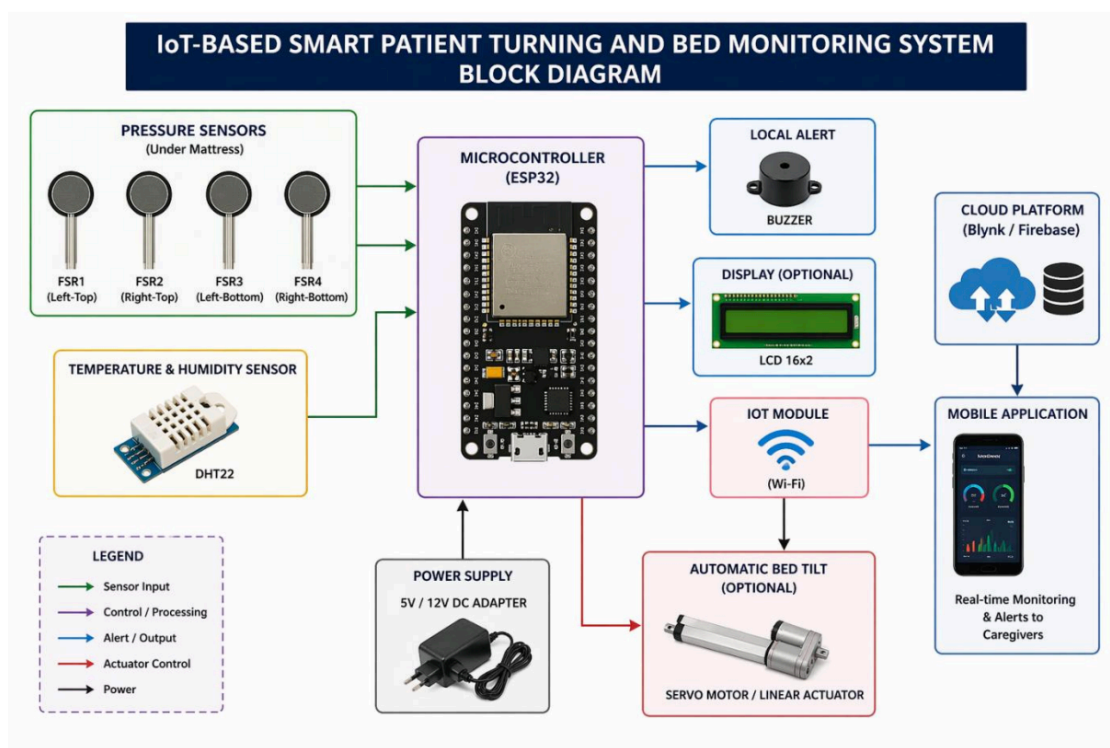
4. Application Layer

- Mobile application/dashboard displays:
 - Pressure values
 - Temperature & humidity
 - Alert status
- Caregivers receive notifications in real-time.

5. Alert & Actuation Layer

- **Buzzer:** Provides local alerts.
- **Optional Servo Motor:** Enables automatic bed tilting for patient repositioning.

7.2 Block Diagram Explanation



As shown in the **block diagram**, the system integrates multiple components into a unified workflow.

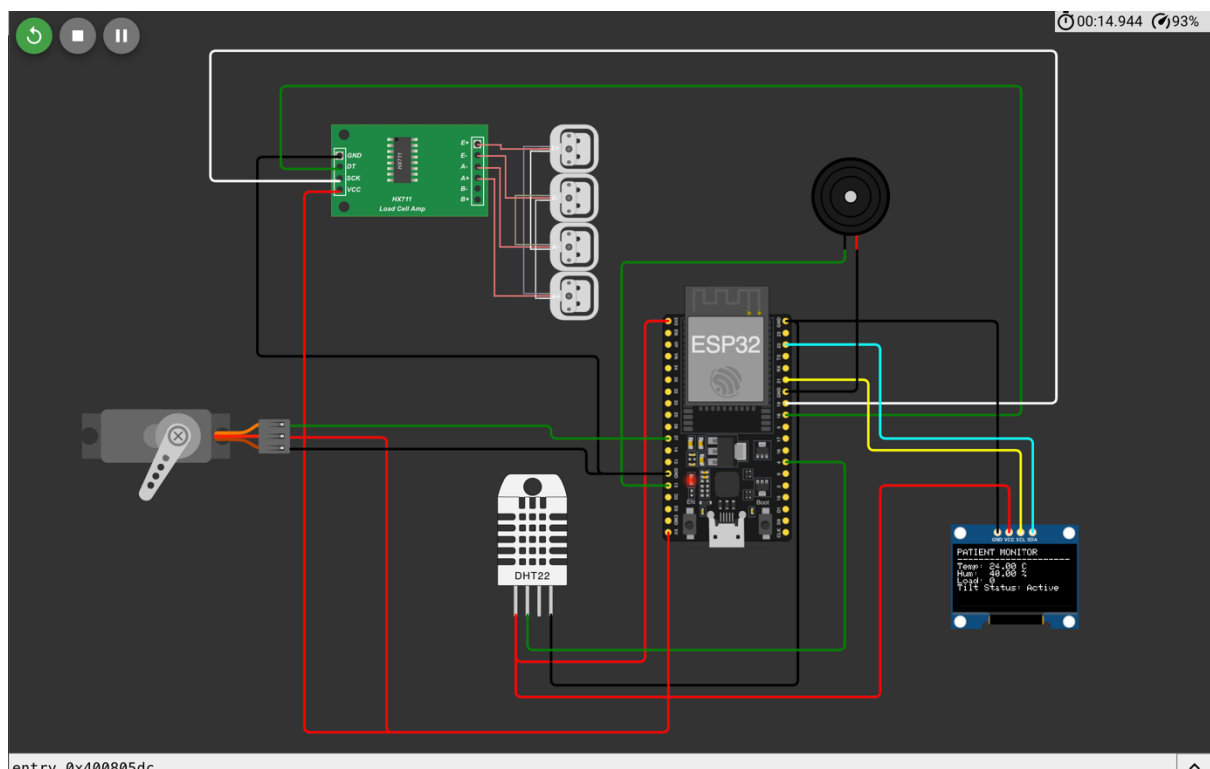
Key Functional Flow:

1. Pressure and environmental data are collected via sensors.
2. Data is sent to ESP32 for processing.
3. Processed data is:
 - Sent to cloud
 - Used for decision-making
4. If abnormal conditions are detected:
 - Buzzer is activated
 - Notification is sent to mobile app

Key Components:

- Input: Pressure sensors, DHT22
- Controller: ESP32
- Output: Buzzer, LCD (optional), Mobile App
- Communication: Wi-Fi module

7.3 Circuit Design



According to the **circuit diagram** , the hardware connections are implemented as follows:

Sensor Connections:

- FSR Sensors → Connected to ESP32 ADC pins (analog input)
- DHT22 → Connected to digital GPIO pin

Output Connections:

- Buzzer → Connected to GPIO output pin
- LCD Display (optional) → Connected via I2C interface

Power Supply:

- ESP32 powered by **5V adapter**
 - Sensors operate at **3.3V**
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7.4 Data Flow Design

The system follows a continuous real-time data flow:

Step-by-Step Data Flow:

1. Sensors collect pressure, temperature, and humidity data.
 2. ESP32 reads sensor values at regular intervals.
 3. Data is processed and analyzed.
 4. Decision logic checks:
 - If patient is immobile for threshold time (e.g., 2–3 hours)
 5. If condition is met:
 - Alert is triggered
 6. Data is uploaded to cloud
 7. Mobile app displays real-time updates
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7.5 Algorithm Design

The system uses a rule-based algorithm for monitoring:

Algorithm Steps:

1. Initialize sensors and Wi-Fi connection
2. Continuously read sensor values
3. Calculate pressure distribution
4. Track time duration of same position
5. IF (time > threshold):
 - Trigger buzzer
 - Send alert notification

6. Upload data to cloud
 7. Repeat loop
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7.6 Database Design (Cloud Storage)

The cloud database (Firebase/Blynk) stores:

Data Fields:

- Patient ID
- Timestamp
- Pressure values (multiple sensors)
- Temperature
- Humidity
- Alert status

Features:

- Real-time synchronization
 - Historical data tracking
 - Remote access
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7.7 User Interface Design

The mobile application provides:

Dashboard Features:

- Live pressure monitoring
- Temperature & humidity display
- Alert notification system
- Patient status (Normal / Alert)

UI Characteristics:

- Simple and user-friendly
 - Real-time updates
 - Accessible for caregivers
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7.8 Power Management Design

- System uses **5V/12V regulated power supply**
 - ESP32 regulates to 3.3V internally
 - Efficient power usage ensures:
 - Continuous monitoring
 - Low energy consumption
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7.9 System Constraints & Design Considerations

- Real-time processing requirement
 - Sensor accuracy and calibration
 - Stable internet connectivity
 - Low-cost hardware design
 - Patient safety and reliability
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Summary of System Design

The system design integrates:

- Hardware (sensors + ESP32)
- Software (algorithm + cloud)
- Communication (Wi-Fi)
- User interface (mobile app)

This ensures a **complete IoT-based healthcare monitoring solution** that is:

- Real-time
- Automated
- Scalable
- Cost-effective